

MAHEEP MOULI SHASHI

ARCHITECT, COMPUTATIONAL DESIGNER & BIM SPECIALIST

Avinguda de Francesc Botey, 51., Barcelona, Spain | maheep.mouli.shashi@gmail.com | +34 692673542 | <https://www.linkedin.com/in/maheep-mouli-shashi-280832180/>



PROFILE

Architect & Computational Designer with 3+ years of experience specialising in **AI-driven design, BIM workflows, and computational methods** for architecture and urban planning. Skilled in integrating **parametric modelling, data analytics, and automation** to deliver innovative, sustainable, and efficient solutions. Adept at working across multidisciplinary teams to optimise design, performance, and execution in complex projects.

EDUCATION

Master in Advanced Architecture (MAA) September 2023 — June 2025

Institute for Advanced Architecture of Catalonia, Barcelona

- Activities and societies: – Studio Research Cluster: Computational Design. Public Review Panel Presenter & Material Systems Lab (Bioplastics + Robotics).
- Completed my **Master's in Advanced Architecture at IAAC** with a **specialisation in Algorithmic and Computational Design**. My academic focus centred on the intersection of urban mobility, data-driven design, and AI in architecture.

Bachelor of Architecture (B.Arch) August 2018 — June 2023

Mysore School of Architecture, Mysore

- Activities and societies: Participated in multiple design competitions and studio projects exploring **urban informality, heritage preservation, and typology adaptation**.
- Completed a **5-year B.Arch** with a strong foundation in **spatial design, construction systems, and architectural theory**.

HARD SKILLS

Rhino and Grasshopper	Project & Workflow Management	BIM	Fabrication & Prototyping
GIS & Urban Analysis	Revit	AutoCAD	Rendering & Visualization
Construction & Management	Python	Naviswork	SketchUp
Adobe Suite	Graph ML	Photography	

SOFT SKILLS

Collaboration & Teamwork	Problem-Solving	Critical Thinking	Creativity & Innovation
Adaptability	Time Management		

PROFESSIONAL EXPERIENCE

Architectural Designer, Atelier 24 March 2023 — May 2025

- Led Rhino-Grasshopper scripting for form generation, reducing modelling time by **40%**. Contributed to the design/documentation of **6,500+ sq. mts.** across hospitals and hotels.
- Produced **200+ technical sheets (floor plans, sections, service layouts)** using **Revit & AutoCAD**.
- Coordinated weekly meetings with MEP teams, boosting issue resolution by 30%. Managed version control systems and drawing repositories across 2 time zones.
- Transformed **client visions into stunning architectural designs**.
- Increased **project efficiency by 30%** through streamlined processes.

Thesis Researcher – Master's Project, Institute for Advanced Architecture of Catalonia (IAAC), Barcelona September 2024 — June 2025

- Built an AI-based congestion index with **82% accuracy in traffic prediction**.
- Visualised city-wide urban congestion using Streamlit & Kepler.gl dashboards. Analysed **5M+ data points from 8 global cities**, integrating **weather, AQI, transit, and population**.

- Compared **AI-managed vs. conventional traffic systems** to model future scenarios. Paper in progress for peer-reviewed publication in urban mobility.
- Led projects from concept to completion, ensuring timely delivery.
- Utilised cutting-edge software for **precise design and visualisation**.

R&D Specialist – Circular Wood Project, WOOD-ID, IAAC, Barcelona

September 2023 — June 2024

- Developed **ML tools for circular wood detection, nesting, and optimisation**.
- Improved fabrication efficiency by integrating CNC/robotic constraints into the design pipeline.
- **Prototyped real outputs using robotic arms and laser cutters**.
- Presented at IAAC digital fabrication and sustainability reviews.
- Designed a **modular timber structure** using **parametric modelling in Grasshopper** to reduce offcut waste by **22%** in fabrication simulations.
- Coordinated with the fabrication lab to generate **CNC-ready cut files** from parametric definitions, decreasing prep time from 6 hours to **under 2 hours**.

Volunteer Assistant, B.A.A.R. Studio – Building with Art, Architecture, Vienna

July 2025 — Present

- Assisting in the **design and development** of community-focused architectural projects.
- Utilised **advanced software tools, including Rhino and Grasshopper**, for parametric **modelling and visualisation**.
- Supported **project management** efforts, ensuring timely delivery and adherence to design specifications.

Architectural Intern, K12 Atelier

August 2022 — February 2023

- Worked on a 4-storey commercial complex from design to site coordination. Supported **redlining, 3D modelling, and presentation decks for client meetings**.
- Assisted in a competition submission shortlisted in the top 10 of 100+ entries.
- **Managed project timelines and deliverables**, ensuring alignment with client expectations and industry standards.

AWARDS AND PUBLICATIONS

Best Project Award, IaaC, Barcelona

AI in Architecture Studio IAAC (2024) – Recognized for the **WOOD-ID** initiative, improving wood efficiency by 25%.

Communique , Urban Village Charitable Trust (UVCT), New Delhi

International Conference on Urban Villages: A Transformational Discourse in Architecture and Planning (22–23 May 2022)

Published by Urban Village Charitable Trust (UVCT), New Delhi.

Paper presented on urban village transformation, highlighting participatory and sustainable planning approaches.

WORK AUTHORIZATION

Eligible to work in Spain/EU

ADDITIONAL INFORMATION

- **Languages:** English, Spanish, Hindi
- **Links:** Linkedin, Portfolio